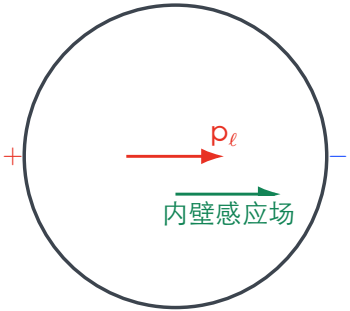
接地圆柱壳内的轴线偶极：正则一次模态使内壁电势归零



$$\varphi = \frac{p_\ell}{2\pi\varepsilon_0}\left(\frac{1}{\rho} - \frac{\rho}{R^2}\right)\cos\phi$$

$$\sigma_{\text{内}}(\phi) = -\frac{p_\ell}{\pi R^2}\cos\phi$$

$\mathsf{E}_{\text{感}}(0) = \dfrac{p_\ell}{2\pi\varepsilon_0 R^2}\mathsf{e}_x$，方向与偶极矩相同。